



Matej Ciglencečki

CTO with over 4 years of experience building a distributed machine learning system that serves tens of thousands of active users. My research interests include **multimodal reasoning** and **3D/4D reconstruction**, with openness to related areas. I am **actively preparing to pursue a PhD in machine learning**, with a **primary goal of publishing in top-tier ML conferences**.

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Education

M.Sc, Data Science – [Faculty of Electrical Engineering and Computing](#), Zagreb, Croatia

Oct 2021 – June 2023

Relevant courses: Machine Learning, Deep Learning, Deep Generative Modeling, Multivariate Data Analysis, Signal Processing, Natural Language Processing, Numerical Mathematics, Advanced Algorithms and Data Structures, Parallel Programming

Master's thesis: [Novel View Synthesis Using Neural Radiance Fields \(NeRF\)](#)

B.Sc, Computer Science – [Faculty of Electrical Engineering and Computing](#), Zagreb, Croatia

Oct 2017 – July 2021

Relevant courses: Algorithms and Data Structures, OOP, Design Patterns, Databases, Probability and Statistics, Artificial Intelligence

Bachelor's thesis: Facial Expression Recognition Using Convolutional Neural Networks

Experience

[TensorPix](#) – CTO

Dec 2024 – present

- Researched and trained image and video super-resolution models (**PyTorch**).
- Developed a custom video inference engine for efficient video processing (**PyTorch**, **Python**, **ONNX**, **TensorRT**).
- Improved super-resolution benchmarking and evaluation suite (**PyTorch**, **Python**).

[TensorPix](#) – Machine Learning Engineer

Nov 2023 – Dec 2024

- Researched and trained video super-resolution and low-light enhancement models (**PyTorch**, **Python**).
- Developed a custom distributed ML inference system (**Redis**, **Python**, **Django**).
- Implemented a custom cloud-agnostic instance autoscaler (**Redis**, **Python**).

[Microblink](#) – Machine Learning Engineer Intern

Feb 2023 – Sep 2023

- Researched and developed Neural Radiance Fields (**PyTorch**).
- Implemented a font image classifier via semi-supervised learning for document verification (**PyTorch**).

[Photomath \(Google\)](#) – Software Engineer Intern

July 2022 – Oct 2022

- Developed cloud services that parse, transform, enrich, and deliver events from Photomath's textbook solutions database (Python, GCP, Datastream, Pub/Sub, Dataflow, Cloud Run, GitHub Actions, Spark).

[Memgraph](#) – Software Engineer

July 2020 – Oct 2021

- Designed and implemented the database schema and backend for Memgraph Cloud. It manages AWS EC2 instances and supports monthly user billing based on user usage (Node.js, TypeScript, Express, Sequelize, PostgreSQL).
- Set up the Elastic Stack for observability on AWS EC2, analyzing application logs in Kibana dashboards via AWS CloudWatch.

Research Experience

Image degradation embedder (TensorPix)

- developed a model that predicts a degradation embedding for a given image (blur, noise, compression, low-light)
- conditioned the degradation embedding to different architectures (NAFNet^[Lichen2022], Latent Denoising U-Net^[Frombach2021])
- enables temporally consistent degradation correction, decouples degradation recognition from degradation correction

Image restoration dataset pruning (TensorPix)

- researched and developed a method for pruning image super-resolution and restoration datasets
- reduced LPIPS by ~9% on the real-world test distribution for two separate super-resolution model architectures
- methodologies: ARNIQA^[Agnolucci2023], IC9600^[Feng2022], JPEG Blockiness^[Bhardwaj2018]

[Neural Radiance Fields, segmentation and central object cropping](#) (Microblink)

- researched and developed a model that generates images from novel views and segmentation masks via transferred learning
- researched and developed a scene cropping method to isolate the scene's central object, reducing foreground artifacts
- quantified angle extrapolation constraints under which the model generates high-quality images
- methodologies: NeRF^[Mildenhall2020], Mip-NeRF^[Barron2021]

[A Squeeze of LIME, a Pinch of SHAP: Adding the Flavor of Explainability to Sentence-Level Commonsense Validation](#)

(M. Moslavac, R. Grbelja, M. Ciglencečki), Text Analysis and Retrieval 2023 Course Project Reports

- Trained a DeBERTa-based model for the classification of sentence coherence.
- Analyzed the model's explainability using LIME and SHAP, which are model-agnostic, post-hoc methods.

Extracurricular

[Torch JPEG Blockiness Metric – Paper implementation](#)

2025

- Built a PyTorch JPEG blockiness metric to detect low-quality JPEG compression in production image datasets at TensorPix.
- Vectorized the algorithm to support fast batched image scoring on inputs of the same size.
- Validated the implementation against the original paper's implementation.

[Machine Learning competition – Musical Instrument Classification](#)

2023

- **Won 2nd place overall and 1st place in model performance. Led a team of four.**
- Trained PyTorch convolutional and transformer models to label musical instruments from audio signals. Converted audio into spectrogram images using digital signal processing. Improved models with latent-space outlier removal, mixup regularization, class distribution shift mitigation, and ArcFace metric learning. Deployed the trained model as a FastAPI inference service.
- Wrote both the competition report and the technical documentation.
- LUMEN Data Science is the largest machine learning competition in Croatia.

[StyleGAN2 latent projection inversion](#)

2022

- Built latent space tools for W+ projection, reconstruction, and interpolation of videos.
- Trained and evaluated StyleGAN2 and StyleGAN2-ADA across full-data and low-data regimes.
- Built a PyTorch pipeline for FFHQ-style preprocessing, face alignment, and normalization.

[Machine Learning competition – Computer Vision GeoGuessr](#)

2022

- **Won 2nd place in model performance. Led a finalist team of three.**
- Trained PyTorch convolutional models that predict the latitude and longitude of Google Street View images. Projected geographical coordinates, developed class-similarity loss functions, and scraped datasets. Deployed the trained model as a FastAPI inference service.
- Wrote both the competition report and the technical documentation.
- LUMEN Data Science is the largest machine learning competition in Croatia.

[Driver fatigue detection in an EEG-based system – Paper implementation](#)

2022

- Reproduced the paper's baseline and methods. Trained a model that matched the reported performance.
- Analyzed and extracted entropy features from the drivers' EEG data.
- Trained and compared four different models, achieving 1 percentage point higher performance than reported in the paper.

Languages

Language	Speaking	Writing	Listening	Reading
Croatian	native	native	native	native
English	C1	C1	C1	C1
Italian	A1	A1	A1	A1

Skills

Languages:	Python, TypeScript, Bash
Technologies:	Git, Linux, PyTorch, PyTorch Lightning, Docker, OpenCV, NumPy, LaTeX
Other:	Machine Learning, Computer Vision, Generative Models, Distributed Systems

Certifications and Workshops

Certificate in Advanced Rhetoric and Public Speaking (Heureka)	2026
Certificate in Rhetoric and Public Speaking (Heureka)	2025
Attended Croatian Machine Learning Workshop (CMLW)	2024
Certificate in Soft Skills (EESTEC)	2019